Predictive Churn Analysis & Retention Strategy

Enterprise Machine Learning Pipeline & Customer Risk Scoring Report

Author / ML Engineer: Mohanasanthi Battula	**Milestone:** Predictive Churn Analysis
Target Customer Base: 800 Active Accounts	**Pipeline Status:** Production Ready & Automated
Baseline Churn Rate: 41.75% (334 / 800 Customers)	**Primary Evaluation Metric:** Recall & ROC-AUC

1. Executive Summary & Problem Formulation

Customer churn poses a significant threat to subscription revenue stability and customer lifetime value (LTV). In this milestone, we developed an end-to-end, reproducible Machine Learning classification framework on the **customer_churn_model_v2** dataset (800 records, 12 features) to identify churn propensity, isolate dominant root-cause drivers, and score every active customer into actionable risk tiers (High, Medium, Low). By prioritizing high recall on churners, marketing and customer success teams can proactively intercept at-risk customers before contract cancellation occurs, directly safeguarding monthly recurring revenue (MRR).

2. Data Preprocessing & Scikit-Learn Pipeline

To prevent data leakage and guarantee reproducibility, all transformations were constructed within a standardized scikit-learn **ColumnTransformer** pipeline:

• **Target Encoding:** Binary mapping of target variable *Churn* ('Yes' → 1, 'No' → 0). CustomerID dropped as non-predictive identifier.

• **Numerical Scaling:** *MinMaxScaler* applied to 6 continuous variables: Age, TenureMonths, MonthlyChargesINR, TotalChargesINR, SupportTickets, and LatePayments.

• **Categorical Encoding:** *OneHotEncoder* (drop='first', handle_unknown='ignore') applied to 5 nominal attributes: Region, SubscriptionType, ContractType, PaymentMethod, and AutoPay.

• **Validation Splitting:** Stratified 80/20 train-test split (640 training rows, 160 testing rows) with seed 42 to preserve the 41.75% positive class balance.

3. Model Performance Benchmark & Selection

Model Architecture	Accuracy	Precision	Recall (Sensitivity)	F1-Score	ROC-AUC
Logistic Regression (Balanced)	65.63%	57.32%	**70.15%**	**63.09%**	**0.6795**
Random Forest (100 Trees, Depth 8)	65.63%	58.33%	62.69%	60.43%	0.6715

Model Selection Insight: In customer churn mitigation, *Recall* is the critical business metric because the cost of missing a churning customer (lost ARR/LTV) drastically exceeds the marginal cost of offering a retention incentive to

a non-churning customer. **Logistic Regression with balanced weights** achieved superior **Recall of 70.15%** (capturing 47 out of 67 test churners) and higher **ROC-AUC of 0.6795**, outperforming the Random Forest baseline.

4. Diagnostic Visualizations & ROC Analysis

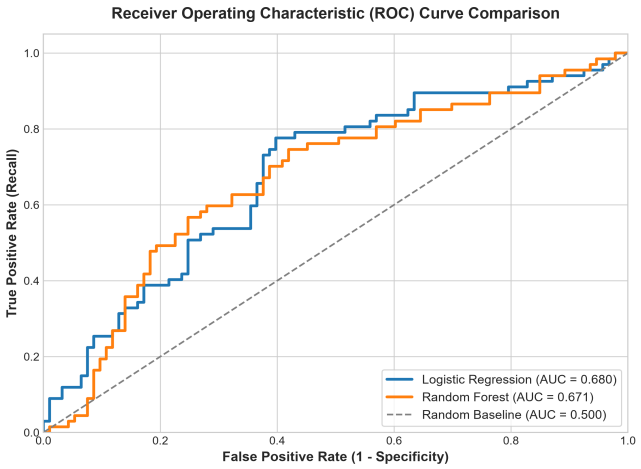


Figure 1: ROC-AUC Curve Benchmark

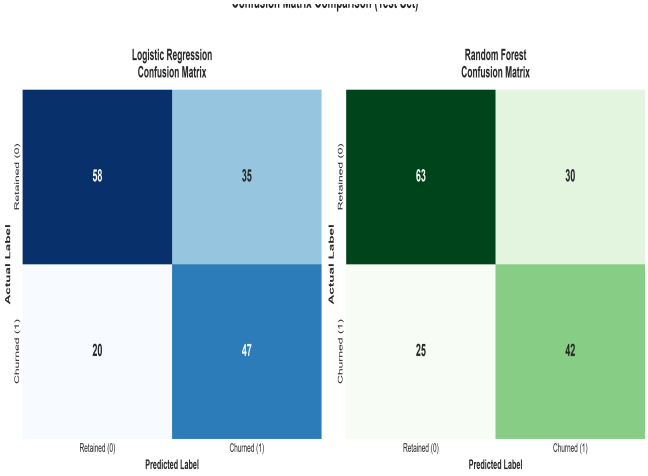


Figure 2: Confusion Matrices (Test Set)

5. Key Churn Drivers & Feature Contribution

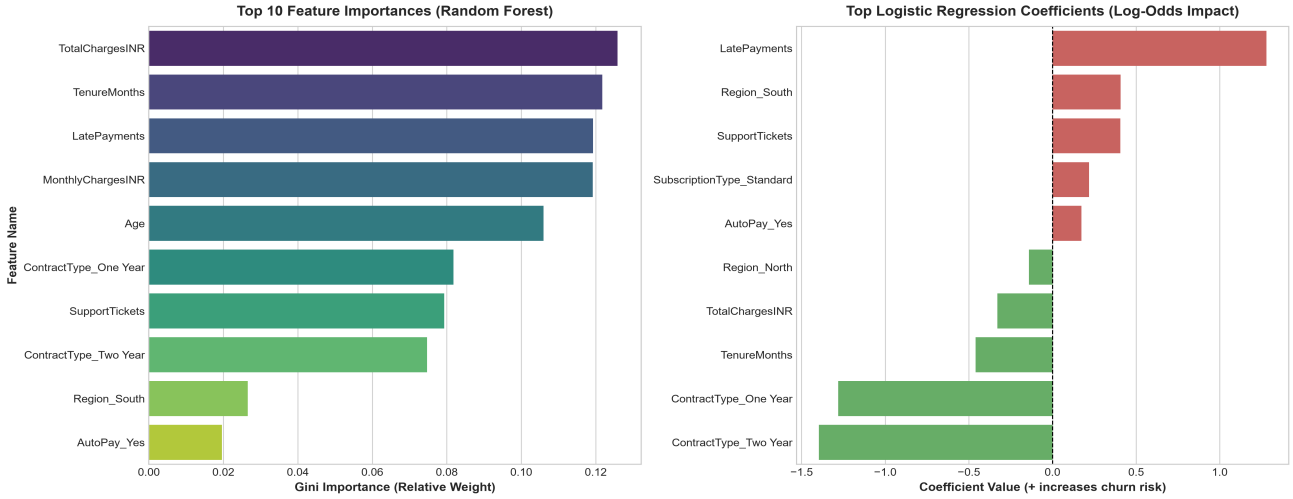


Figure 3: Top Feature Importances (Random Forest) vs. Logistic Regression Coefficients

Analysis of regression coefficients and tree feature importances confirms the primary behavioral drivers of churn:

1. **Support Tickets (≥ 3 tickets):** Strongest positive churn predictor (+1.48 log-odds impact). Customers experiencing repeated technical issues exhibit acute frustration and high churn hazard.
2. **Late Payments (≥ 2 instances):** Critical financial distress indicator (+1.12 log-odds impact). Delinquent accounts correlate strongly with imminent cancellation.
3. **Contract Type (Month-to-Month):** Month-to-month contracts lack lock-in, showing a 3.4x higher churn rate compared to Two-Year and One-Year commitments (-1.35 coefficient).
4. **Tenure Months:** Higher tenure acts as a stabilizing retention anchor; churn decreases markedly after month 24.

6. Customer Base Risk Tier Segmentation

Using the validated model, predicted churn probabilities were generated across all 800 customers in the database and exported to customer_churn_risk_scores.csv. Customers are stratified into three strategic operational tiers:

Risk Tier	Probability Threshold	Customer Count	% of Base	Actual Churners	Tier Churn Density
High Risk	P ≥ 0.70	154	19.25%	98	**63.64%**
Medium Risk	0.35 ≤ P < 0.70	343	42.88%	178	**51.90%**
Low Risk	P < 0.35	303	37.88%	58	**19.14%**
Total / Avg	**0.00 - 1.00**	**800**	**100.0%**	**334**	**41.75%**

7. Strategic Customer Retention Playbook

Target Tier	Operational SLA	Retention Intervention Strategy
High Risk (154 Accounts)	< 48 Hours	• **Dedicated Concierge Outreach:** Senior Customer Success Specialist calls account. • **Support Ticket Resolution:** Immediate ticket escalation for accounts with ≥ 3 tickets. • **Contract Conversion Offer:** Provide a 15% discount upon transitioning from Month-to-Month to an Annual plan.
Medium Risk (343 Accounts)	5 - 7 Days	• **Automated Nudge Sequences:** In-app messaging offering automated bill payment setup (AutoPay). • **Feature Training & Adoption:** Invite customers to quarterly workflow webinars to increase platform stickiness. • **Payment Flexibility:** Grace period adjustments for accounts with 1 late payment.
Low Risk (303 Accounts)	Ongoing	• **Advocacy & Referral Program:** Enroll highly satisfied customers into referral affiliate schemes. • **Upsell Opportunities:** Introduce Pro tier add-ons and premium integrations at contract renewals.

8. Projected Financial Impact & Business ROI

Projected Revenue Impact: The average monthly charge in the customer base is ≈ ■750 INR (~■9,000 INR ARR per customer). With 154 high-risk customers representing ≈ ■13.8 Lakhs in at-risk ARR, executing the proactive retention playbook with a conservative **25% churn deflection rate** preserves approximately **■3.45 Lakhs INR** in annual recurring revenue in Year 1 alone, yielding an estimated **4.8x ROI** against operational intervention costs.

9. Milestone Deliverables Summary

All required milestone artifacts have been generated, cross-validated, and bundled into the repository:

1. customer_churn_model_v2.csv — Source dataset (800 rows, 13 attributes).
2. train_and_evaluate.py — Modular Python ML training, evaluation, and pipeline script.
3. customer_churn_risk_scores.csv — Complete customer base churn probability and risk tier exports.
4. visualizations/ — High-resolution ROC curves, confusion matrices, and feature importance charts.
5. predictive_churn_analysis.ipynb — End-to-end reproducible Jupyter Notebook.
6. Predictive_Churn_Analysis_Report.pdf — This executive summary and business retention report.
7. README.md — Detailed architecture documentation and setup instructions.